



Mo Tu We Th Fr Sa Su

Memo No. _____

Date / /

15. Conversion of NFA to DFA

Every DFA is an NFA, but not vice versa

But there is an equivalent DFA for every NFA

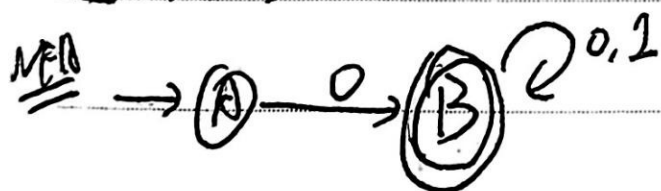
$$\text{DFA } \delta = Q \times \Sigma \rightarrow Q \quad Q \in 2^Q$$

$$\text{NFA, } \delta = Q \times \Sigma \rightarrow 2^Q$$

eg. $\text{NFA} \cong \text{DFA}$

$L = \{ \text{Set of all strings over } (0,1) \text{ that starts with '0'} \}$

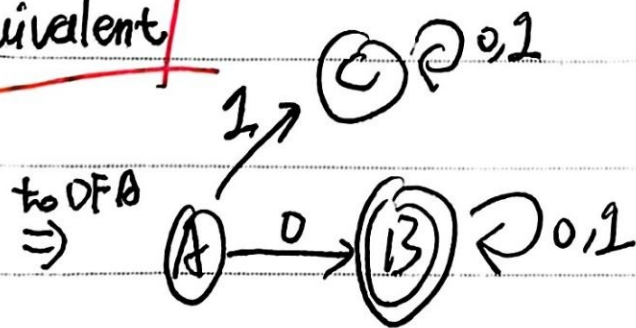
$$\Sigma = \{0,1\}$$



state transition

	0	1
A	B	ϕ
B	B	B

equivalent



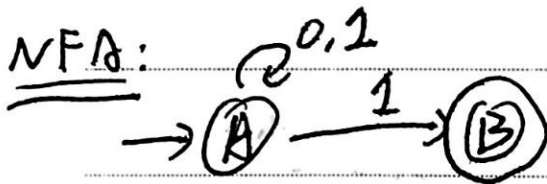
	0	1
A	B	C
B	B	B
C	C	C

C-Dead state/
Trap state

16. NFA TO DFA EX1

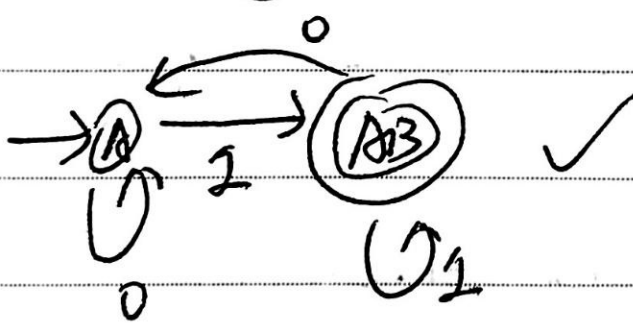
$L = \{ \text{Set of all strings over } \{0, 1\} \text{ that ends with '1'} \}$

$$\Sigma = \{0, 1\}$$



	0	1
A	{A}	{A, B}
B	ϕ	ϕ

DFA diagram



	0	1
A	{A}	<u>AB</u>
AB	<u>{A}</u>	<u>{AB}</u>

{A} $\cup \phi$ {AB} $\cup \phi$

AB is a single state

Subset construction method

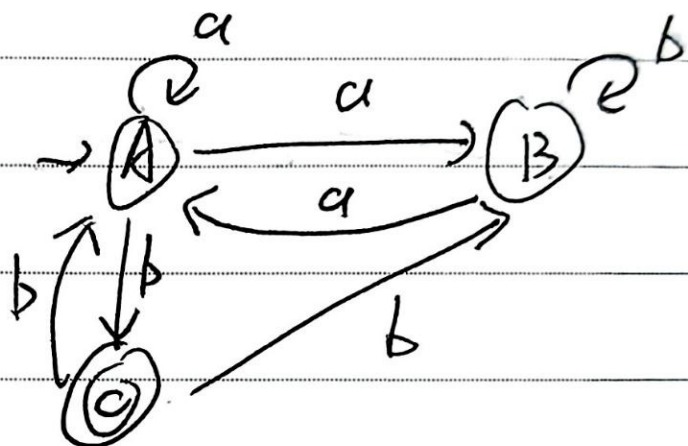


17. NFA To DFA Ex2

eg.

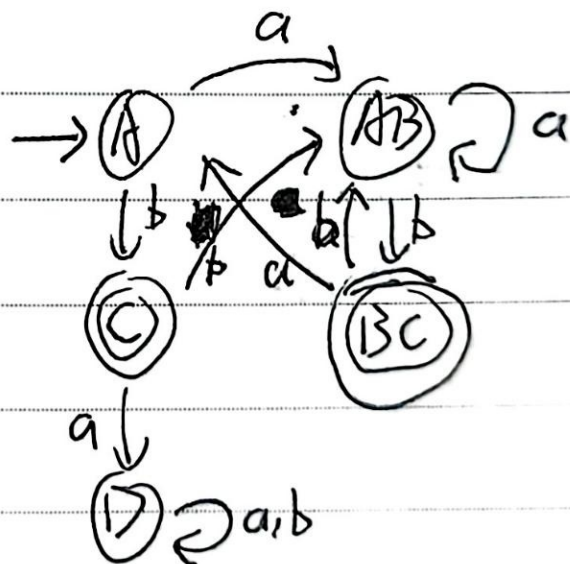
Find the equivalent DFA for the NFA given by $M = [\{A, B, C\}, \{a, b\}, \delta, A, \{C\}]$ where δ is given by:

	a	b
$\rightarrow A$	AB	C
B	A	B
$\odot C$	-	AB



	a	b
$\rightarrow A$	<u>AB</u>	<u>C</u>
AB	AB	<u>BC</u>
*BC	A	AB
$\odot C$	∇	AB
∇	$\uparrow \nabla$	∇

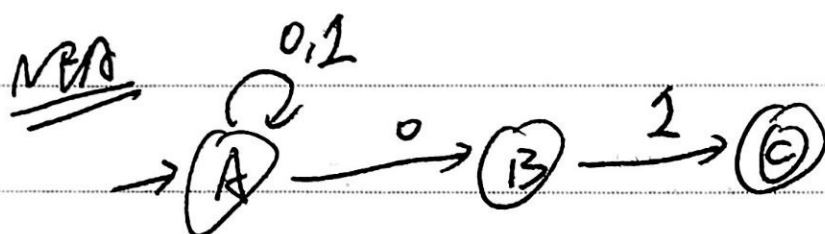
Dead State



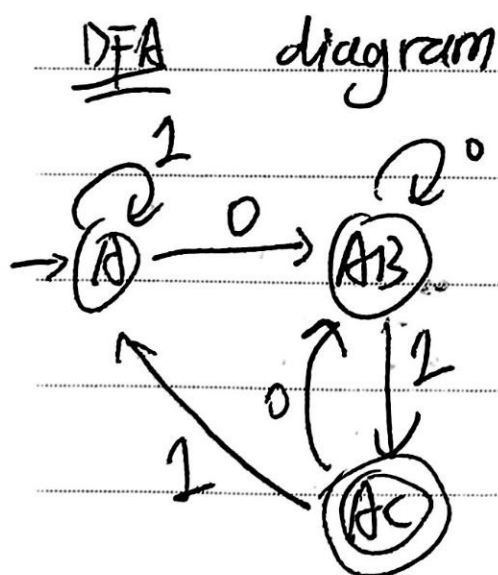
Assignment: $L = \{ \text{all strings over } (a, b) \text{ that ends with 'ab' or just one 'b'} \}$

18. NFA to DFA Example 3.

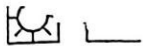
$L = \{ \text{Set of all strings over } \{0,1\} \text{ that ends with '01'} \}$. Construct its equivalent DFA



	0	1
A	AB	A
B	ϕ	C
C	ϕ	ϕ



	0	1
A	<u>AB</u>	A
AB	AB	AC
AC	AB	A

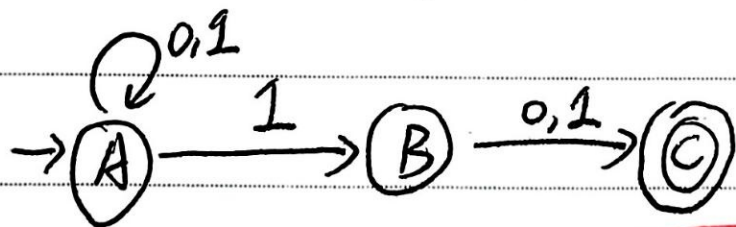
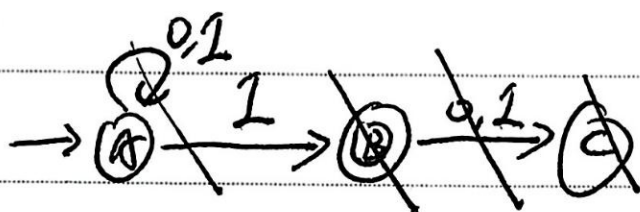


19. Conversion of NFA to DFA - Example

Design An NFA for a language that accepts all strings over $\{0, 1\}$ in which the second last symbol is always '1'. Then convert it to its equivalent DFA.

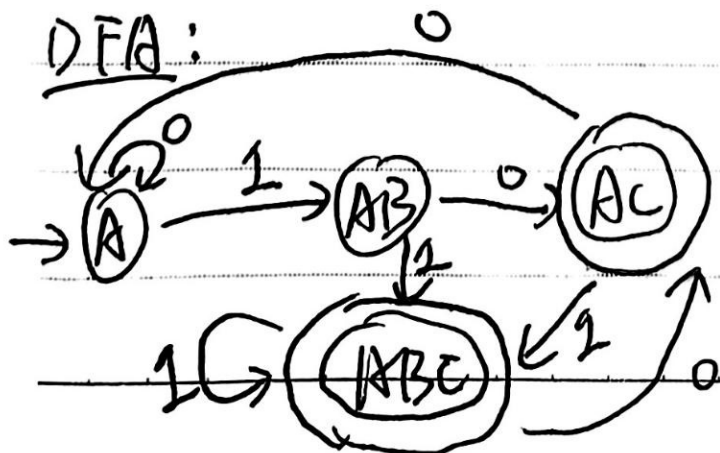
$$\Sigma = \{0, 1\}$$

NFA:



	0	1
A	A	AB
B	C	C
C	ϕ	ϕ

DFA:



	0	1
A	A	AB
AB	AC	ABC
AC	ABC	ABC
ABC	ABC	ABC